

# **Online Examination System**

## **Introduction**

The Online Examination and Performance Analytics System is a full-stack web application developed using Angular and Django. The system is designed to digitize and automate examination processes in educational institutions. It ensures secure online testing, real-time scoring, performance analysis, and ranking classification based on student performance levels.

The platform supports two main roles: Admin and Student. It enables efficient exam management, question bank organization, and analytical reporting through a modern web interface.

## **Objective**

- To develop a secure online examination platform.
- To provide timer-based automated testing.
- To classify questions based on difficulty levels.
- To automatically evaluate and generate results instantly.
- To rank students based on performance (High / Medium / Low).
- To generate suggestions based on scores.

## **Scope of the Project**

- Multi-role platform (Admin and Student)
- Online exam scheduling and management
- Question bank with difficulty categorization
- Timer-based exam system
- Automatic result calculation
- Ranking and grading mechanism

## **Tech Stack**

### **Backend**

- Python 3.x – Core programming language
- Django – Web framework

- Django REST Framework – API development
- SQLite / PostgreSQL – Database
- JWT Authentication – Secure login system

## **Frontend**

- Angular – Frontend framework
- HTML5 / CSS3 – UI design
- Bootstrap – Responsive design
- JavaScript – Client-side scripting

## **Modules Description**

### **1. Admin Module**

#### **a. User Management**

- i. Create and manage student accounts
- ii. View registered students
- iii. Block or remove users

#### **b. Exam Management**

- i. Create new exams
- ii. Set exam duration
- iii. Schedule exam date and time
- iv. Assign total marks

#### **c. Question Bank Management**

- i. Add questions
- ii. Set difficulty level (Easy / Medium / Hard)
- iii. Edit or delete questions
- iv. Assign questions to exams

#### **d. Results Monitoring**

- i. View all student results
- ii. Filter by exam
- iii. View ranking levels

#### **e. Suggestion Generator- Based on student percentage:**

- i. Above 75% → Excellent performance, attempt advanced exams

- ii. 40%–75% → Good performance, practice medium/hard questions
- iii. Below 40% → Needs improvement, focus on basics

**f. Analytics Dashboard**

- i. Total Students
- ii. Total Exams Conducted
- iii. Average Score
- iv. High / Medium / Low Performance Count
- v. Graphical Reports

**2. Student Module**

**a. Authentication**

- i. Secure login
- ii. View scheduled exams

**b. Question Bank**

- i. View sample questions
- ii. Practice difficulty-based questions

**c. Timer-Based Test**

- i. Countdown timer during exam
- ii. Automatic submission when time expires

**d. Automatic Scoring**

- i. Backend verifies answers
- ii. Score calculated instantly
- iii. Percentage generated automatically

**e. Ranking System**

- i. HIGH – 75% and above
- ii. MEDIUM – 40% to 74%
- iii. LOW – Below 40%

**f. Result Dashboard**

- i. Exam Score
- ii. Percentage
- iii. Ranking Level
- iv. Suggestions
- v. Past Exam History

**g. Student Analytics**

- i. Improvement tracking
- ii. Difficulty-wise performance analysis

## **Existing System**

- Manual paper-based examinations
- Time-consuming result processing
- High risk of human errors
- No centralized analytics
- No instant feedback system
- No ranking classification

## **Proposed System**

- Fully automated online examination system
- Timer-based secure testing
- Instant automatic scoring
- Difficulty-level classification
- Ranking system (High / Medium / Low)
- Intelligent performance suggestions
- Real-time analytics dashboards
- Secure role-based authentication

## **Advantages**

- Reduces manual workload
- Faster result generation
- Transparent evaluation process
- Easy exam scheduling
- Performance tracking and analytics
- Secure and scalable architecture

## **Future Enhancements**

- AI-based question generation

- Adaptive testing system
- Email result notifications
- Certificate generation
- Mobile application integration
- Proctoring system with webcam monitoring